AHMED ALVEE

Solutions Architect | AI & Automation Engineering
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Summary

Solutions architect with a background across robotics research, C++ medtech software, healthcare data integration on AWS and LLM workflow design. Built AWS data pipelines and LLM workflows, took them live with customers on site, led the first overseas deployment at Mesh Bio and led a team of 3 solutions engineers. Designed an AI monitoring workflow for Truvalu as part of a 5-person MBA team in the RSM consulting project. AWS Certified Solutions Architect – Associate. Completing an MBA at Rotterdam School of Management.

Skills

Cloud: AWS (Lambda, Step Functions, SQS, SNS, EventBridge), Terraform
AI and LLM: Model Context Protocol (MCP), Amazon Bedrock, LangGraph, LangChain, Gemini, Pydantic, ChromaDB, n8n, Claude Code, Cursor
Integration: REST APIs, Postman, HL7, MLLP, DICOM, Microsoft Graph API
Languages and tools: Python, C++, MongoDB, Git, Linux, Qt, ROS
Ways of working: Agile teams, sprints, Scrum, stakeholder management

Professional Experience

Mesh Bio Pte Ltd — Singapore, SG
Solutions Architect (prev. Senior Solutions Engineer) — 05/2022 – 12/2025
Singapore health-tech company working with private healthcare providers. Its most widely adopted product was a health screening workflow that scored each patient's risk and built screening reports. Built the pipelines behind it.

- Per-submission time improved from about 1 hour to 6-10 minutes, including human review, for 30+ clinics, by building an AI clean-up pipeline from scratch (LangGraph, Claude on Bedrock, Pydantic, ChromaDB) in which a human reviewer checked and submitted every file.
- Developed a configurable Pydantic validation layer, live for the APIs and the lab and radiology pipelines, and set strict criteria for submissions as a responsible AI control.
- Built proofs of concept and MVPs, ran pilots with teams in Indonesia before wide deployment, and extended the pipeline to run on premises (not cloud) as an MVP for a pilot with an Indonesian client.
- Onboarding time improved from 3-5 days to under 1 day for each new lab and clinic combination, by redesigning a hardcoded Lambda per combination, which had meant copying code and testing incoming data against it, into a config-driven AWS pipeline (Lambda, SNS, SQS, EventBridge Pipes, Step Functions, HL7 integration) that handled about 10,000 records an hour in Singapore and stayed inside Lambda and MongoDB connection limits.
- Protected patient profiles from wrong data with record count checks, per-record isolation and exception handling that stopped bad runs, so a lab vendor's unannounced system switch was resolved in one day with a solutions engineer.
- Built the MLLP endpoint that receives HL7 from lab vendors and writes files to S3, and handled hourly scheduled files plus on-demand submissions for the lab, radiology and clinic pipelines.
- Turned the company's first overseas customer into a lighthouse account that opened follow-on regional projects, by recommending a headless API-based workflow, cutting scope to meet the deadline, and leading 3 weeks of on-the-ground deployment.

- Joined technical sales calls to map customer tech stacks and integration points, and owned the integration APIs, built ahead of demand with Postman documentation, so fewer interfaces had to be built reactively.
- Managed internal and external stakeholders as Solutions Engineer and then Solutions Architect, working in Agile teams with sprints and Scrum.
- Promoted to Solutions Architect; hired (2024 to mid 2025), trained and supervised a team of 3 solutions engineers as they took over and expanded the APIs and pipelines.

Creative Medtech Solutions Pte Ltd — Singapore, SG
Software Engineer — 09/2021 – 04/2022
Singapore medtech company building a robotic system that ablates cancer cells with high intensity focused ultrasound (HIFU).

- Owned the annotation and user-interaction layer of the imaging system for a HIFU cancer-ablation robot: marking and measuring ablation targets, and DICOM export.

Singapore University of Technology and Design — Singapore, SG
Research Assistant — 02/2018 – 08/2021
University research lab building robotics and wearable systems, working with teams of research assistants and PhD students.

- Mapped 2 floors and 5 rooms in under 6 minutes with a 12+ robot system, owning its locomotion code.
- Recognized 7 hand gestures to command unmanned vehicles with an ML pipeline (Python, ROS, StretchSense).
- Contributed C++ and hardware engineering to 2 more robotics builds: a legged mapping robot and a custom PCB for 3D-knitted wearables.

MBA Consulting Project

RSM Living Management Project — Rotterdam, Netherlands
AI Strategy Consultant (Client: Truvalu) — 06/2026 – 07/2026
MBA consulting project for Truvalu, a Netherlands-based impact investor supporting agri-food SMEs in emerging markets.

- Monitoring time for an impact investor was projected to improve by about 45% with a 5-stage AI monitoring workflow, designed as part of a 5-person MBA team.
- Engineered and demoed AI tools and agents (Gemini, Claude, Tally.so) as proofs of concept and an MVP spanning all 5 workflow stages, from data clean-up to reporting.
- Co-led the final client presentation to Truvalu leadership.

Education

Rotterdam School of Management, Erasmus University — Rotterdam, Netherlands
Master of Business Administration (MBA) — 01/2026 – 03/2027

Singapore University of Technology and Design — Singapore, SG
BEng (Honours), Engineering Product Development — 05/2014 – 08/2017

Certifications

- AWS Certified Solutions Architect – Associate, Amazon Web Services (10/2023).